The Selling closure: complete reindexing of measured lattices

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Abstract

Two measurements of one crystal can reduce to different cells of the same lattice, and the integer matrix between them need not be a lattice symmetry; a reindexing that searches only symmetries, or only a list of small matrices, can miss it. This paper takes all Selling-reduced cells (obtuse superbases) of a lattice, with the unimodular matrices between them, as one object, the Selling closure. It is a finite groupoid whose arrows at a cell are the lattice point group and whose arrows between two settings form a coset of it; every arrow belongs to a fixed set of 6096 integer matrices, and the closures of Delone's 24 sorts are computed from those matrices alone. For a measured cell the closure is widened to every superbase within a strain ϵ of obtuse, in the affine-invariant norm on metric tensors, which is the same in every basis. The widened set is finite and is enumerated exactly; at the bands used here it lies in a first layer of 127 superbases for the cells of ordinary shape tried and in a second layer for needles and plates. Reindexing a new cell onto a reference then needs one reduction of the new

cell, and every unimodular operator whose price is at most ϵ is among the candidates. The candidates form a coset of the band's pseudo-symmetry group, their classes under the Laue group are the distinct indexings, and a pseudo-twofold enters at twice its distance to the lattice it would fix. On 780 constructed cases the true operator was always found; a search over small integer matrices between the reduced cells finds it in all but two, so the gain is the guarantee and the structure of the candidate set rather than the hit rate.

1. Introduction

Two questions recur in indexing and data processing: what are the symmetries of a lattice, and are two cells the same lattice, and if so by which integer matrix? Both are usually answered by searching. Symmetry is found by scanning for near-twofolds within an angular tolerance (Le Page, 1982; Sauter *et al.*, 2004); reindexing tries a list of small integer matrices, or the operators of the point group just found. The symmetry search is complete within its index bound. The reindexing search is complete only if the list happens to contain the operator. Both are complicated by the same fact: a lattice with a right angle has more than one reduced cell, and the alternatives are related by integer matrices that are not symmetries.

Selling reduction makes this multiplicity explicit. A superbase is four lattice vectors summing to zero; it is obtuse, or Selling-reduced, when all six pairwise dot products are non-positive (Selling, 1874). A superbase is obtuse exactly when its partial sums are the Voronoi vectors of the lattice (Voronoi, 1908; Conway & Sloane, 1992), the obtuse superbase is unique except when a dot product vanishes (Delaunay, 1933; Patterson & Love, 1957), and Kurlin (2022) listed all obtuse superbases of each zero pattern and showed that the six values, arranged in their three pairs of opposite slots, form a complete isometry invariant of the lattice, the root invariant. Delaunay's 24 sorts

classify lattices by the zeros and equalities among those values (Delone *et al.*, 1975; Burzlaff & Zimmermann, 2006; Andrews & Bernstein, 2023). The same objects appear as S^6 with its boundary transformations (Andrews *et al.*, 2019*a*,*b*) and in SELLA (Andrews *et al.*, 2023). On the Niggli side, the multiplicity of Buerger cells (Buerger, 1960; Gruber, 1973) and the instability of the reduced cell under error (Andrews *et al.*, 1980; Andrews & Bernstein, 1988) are the same phenomenon. For measured cells, distances to Bravais-lattice subspaces are measured by Andrews & Bernstein (2014) and by Oishi-Tomiyasu (2012, 2024), whose error-stable Bravais-lattice determination comes with a bound under which a finite list of basis transforms is provably enough; the operators relating two settings form a coset of the symmetries (Flack, 1987).

This paper takes the whole set of obtuse superbases of a lattice, with the integer matrices between them, as the object of study. We call it the *Selling closure*. What is added to the classical material is:

1. the closure as a finite groupoid, holding the point group and every alternative reduced setting, with the counts of visible symmetries and cell shapes for all 24 sorts computed from a universal set of integer matrices alone;

2. for a measured cell, the widened closure in a strain norm that is the same in every basis, a finite set that is enumerated exactly;

3. its description by layers of that universal set, exact up to an adjacency radius computed for each cell: the first layer for the cells of ordinary shape tried, a second layer for needles and plates;

4. reindexing from one reduced cell of the new measurement, with a guarantee that every operator within the band is among the candidates, the candidates forming a coset whose classes under the Laue group are the distinct indexings, and a rule for the price at which a pseudo-symmetry is offered.

The reindexing guarantee, not the hit rate, is the practical gain: on constructed cases a search over small integer matrices between the two reduced cells finds the true operator almost as often (Section 5.4).

Section 2 collects the classical definitions. Section 3 defines the closure as a groupoid and works through one monoclinic cell. Section 4 states the computations for measured cells. Section 5 gives the closures of the 24 sorts, the distances of measured cells to the sorts, the completeness of the computation and the reindexing results. Proofs are in Appendix A and the simpler theory of an absolute band on the conorms in Appendix B; the catalogues of shapes and specialisations, the construction of the operator set with no lattice, and the list of computational checks are supporting information (Sections S1–S4). Every computed number in the text carries a mark [z*nn*] naming the check that reproduces it, placed at the end of the sentence it supports; the checks are test files in the `agentsg` repository, listed in the supporting information, and [lib] marks the library's own test file.

2. Background

2.1. Superbases, conorms and the metric tensor

Selling reduction works with four vectors: the cell vectors a, b, c and $d = -a-b-c$. The four form a *superbase*; any three of them are a basis of the lattice. The six negated pair products

$$p_{da} = -d\cdot a, \quad p_{db} = -d\cdot b, \quad p_{dc} = -d\cdot c, \quad p_{ab} = -a\cdot b, \quad p_{ac} = -a\cdot c, \quad p_{bc} = -b\cdot c$$

are the *conorms* (Selling, 1874; Conway & Sloane, 1992); the Selling parameters of S^6 (Andrews *et al.*, 2019*a*,*b*) are the same numbers with the opposite sign. Conorm vectors are written in the slot order da, db, dc, ab, ac, bc, except that the sort patterns of Table 4 use the S^6 order bc, ac, ab, ad, bd, cd for comparison with Andrews & Bernstein (2023). A superbase is *obtuse*, which is to say Selling-reduced, when all six conorms

are non-negative. The metric tensor in the basis (a, b, c) is linear in the conorms,

$$\mathbf{G} = \begin{pmatrix} p_{ab} + p_{ac} + p_{da} & -p_{ab} & -p_{ac} \\ -p_{ab} & p_{ab} + p_{bc} + p_{db} & -p_{bc} \\ -p_{ac} & -p_{bc} & p_{ac} + p_{bc} + p_{dc} \end{pmatrix}, \qquad (1)$$

so the conorms are six cell parameters. A zero conorm is a right angle between two of the four vectors; the set of slots that vanish is the *zero pattern*. If the non-zero conorms are to connect the four vectors, there are five kinds of zero pattern up to relabelling, which correspond to the five types of Voronoi polyhedron (Voronoi, 1908; Conway & Sloane, 1992); with the two placements of three zeros (a triangle or a path) there are 38 faces of the cone of non-negative conorms, a face being where a chosen set of conorms vanishes.

2.2. Flips and swaps

Two kinds of integer operator act on a superbase.

Flips. A flip at the conorm p_{ab} replaces a by $-a$ and adds the old a to the two other vectors, $(a, b, c, d) \mapsto (-a,\ b,\ c + a,\ d + a)$. The new conorms are $p'_{ab} = -p_{ab}$, $p'_{ac} = p_{da} + p_{ab}$, $p'_{da} = p_{ac} + p_{ab}$, $p'_{bc} = p_{bc} + p_{ab}$, $p'_{db} = p_{db} + p_{ab}$ and $p'_{dc} = p_{dc} - p_{ab}$; this is the boundary transformation of Andrews *et al.* (2019*a*), written on the basis. Selling reduction applies flips at negative conorms until none remain. When p_{ab} is exactly zero the flip exchanges the slots ac and da and carries one obtuse superbase to another of the same lattice, which is why the reduced cell of a lattice with a right angle is not unique; all reduced cells of a lattice carry the same six conorm values, from which the root invariant of Kurlin (2022) is formed. In the basis (a, b, c) each flip is a fixed 3×3 integer matrix of determinant -1 that is its own inverse (Table 1). A word of flips such as *ab.bc.db* is applied left to right, each flip read in the current labelling, and its matrix is $\mathbf{F}_{ab}\mathbf{F}_{bc}\mathbf{F}_{db}$, columns being the new basis vectors in the old basis.

Table 1. *The six Selling flips as* 3×3 *integer matrices in the basis* (a, b, c), $d = -a - b - c$; *columns are the new basis vectors in the old basis. Each has determinant* -1 *and squares to the identity. For the flips at da, db, dc the vector* d *is negated and the new* a, b, c *are listed.*

flip at	new basis	matrix
ab	$(-a,\ b,\ a+c)$	$\begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
ac	$(-a,\ a+b,\ c)$	$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
bc	$(a+b,\ -b,\ c)$	$\begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
da	$(a,\ -a-c,\ -a-b)$	$\begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$
db	$(-b-c,\ b,\ -a-b)$	$\begin{pmatrix} 0 & 0 & -1 \\ -1 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix}$
dc	$(-b-c,\ -a-c,\ c)$	$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix}$

Swaps. A superbase is an unordered set of four vectors up to a common sign, so one reduced cell can be written in $4! \times 2 = 48$ ways. The 48 reorderings and sign changes are the swaps; they change which vectors are called a, b, c and nothing else. A swap is written by listing which vectors become the new d, a, b, c, with a leading sign for the common sign change: $-(a, c, b, d)$ makes the old a the new d, the old c the new a, keeps b, makes the old d the new c, and negates all four.

2.3. Delone sorts

Delaunay (1933; the same author is cited as Delone from his Russian-language work) classified lattices into 24 sorts by the pattern of zeros and equalities among the reduced conorms. Each sort lies in one Bravais type, and some Bravais types comprise several sorts (Delone *et al.*, 1975; Burzlaff & Zimmermann, 1985, 2006). We use the patterns and symbols C1 to A3 of Andrews & Bernstein (2023) and SELLA (Andrews *et al.*, 2023). The sorts are the Selling counterpart of the 44 Niggli lattice characters

(Niggli, 1928; de Wolff & Gruber, 1991): they split a Bravais type by the shape of the reduced cell, not by its symmetry. A zero conorm restricts the metric without adding a symmetry: one right angle between two superbase vectors gives no twofold, so the triclinic sorts A2 and A3, with one and two zeros, are as triclinic as A1.

2.4. Terms

Table 2 collects the terms used in the rest of the paper, with the section in which each is defined.

Table 2. *Terms and symbols.*

term	meaning	defined
superbase, conorm	four vectors summing to zero; the six negated pair products p_{ij}	2.1
obtuse, reduced cell	all six conorms ≥ 0; an obtuse superbase, in some labelling	2.1
zero pattern, face	the slots whose conorms vanish; the set of conorm vectors with those zeros, in the cone $p \geq 0$	2.1
flip, swap	the integer operators of Table 1; the 48 relabellings of a superbase	2.2
sort	one of Delone's 24 classes of lattices by zeros and equalities among the conorms	2.3
object, arrow	a labelled obtuse superbase (a metric tensor $\mathbf{G}$); a unimodular $\mathbf{P}$ with $\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}'$	3.1
visible symmetry, shape	a symmetry that relabels the reduced cell; the orbit of a superbase under the point group	3.1
$\mathcal{U}$, layer $\mathcal{N}_k$	the 6096 arrows of all closures; $\mathcal{N}_1 = \mathcal{U}F_0$, $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$	3.3, 4.3
strain, equation (2)	$\|\mathbf{G}^{-1/2}\Delta\mathbf{G}\mathbf{G}^{-1/2}\|_F$, the same in every basis	4.1
band ϵ	the strain within which a superbase, or an operator, is accepted	4.2
depth, price	strain from the measured metric to the nearest one that makes a superbase obtuse; strain by which an operator fails to carry one metric onto another	4.2
widened closure $\mathcal{S}_\epsilon$, H_ϵ	the superbases of depth $\leq \epsilon$; the self-maps of the cell of price $\leq \epsilon$	4.2
adjacency radius δ_k	the strain at which a superbase outside $\mathcal{N}_k$ first becomes obtuse	4.3
sort spectrum, gap	the 24 distances of a cell to the sorts; the largest ratio between consecutive distances	4.4, 5.2
indexing class	a class of the Laue group among the candidate operators	4.5
difference-wall member	a member of the widened set reachable from the reduced cell only through a non-member	5.3
λ_1, parity class	the squared length of the shortest lattice vector; a class of $\Lambda/2\Lambda$	App. A, B

3. The Selling closure as a groupoid

3.1. Definition

A *groupoid* is a collection of objects with invertible arrows between them, where two arrows can be composed, one applied after the other, only when the first ends where the second starts (Brandt, 1927; Higgins, 1971). A lattice has no single preferred setting, other than by convention; it has a finite set of obtuse superbases, and between them there are two kinds of arrow. A symmetry maps a setting to itself and leaves its metric tensor unchanged. A reindexing maps one setting to another: the metric tensor is rewritten in the new basis, but the lattice is untouched and no strain is induced. A group captures only the first kind; the groupoid holds the point group and every equivalent setting in one structure.

A *labelled* obtuse superbase is an obtuse superbase together with a choice of which three of its vectors, in which order and with which common sign, are called a, b, c; each obtuse superbase has 48 labellings, joined by swaps. The *Selling closure* $\mathcal{C}(\mathbf{G})$ of a lattice is the groupoid whose objects are the metric tensors of the lattice in its labelled obtuse bases, and whose arrows from $\mathbf{G}$ to $\mathbf{G}'$ are the unimodular matrices $\mathbf{P}$ with $\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}'$: the action groupoid of $GL_3(\mathbb{Z})$ on those metric tensors. Two labelled superbases related by a symmetry give the same metric tensor, so there are $48n/|\text{point group}|$ objects when the lattice has n obtuse superbases. In counting, the rest of the paper speaks of the n unlabelled *superbases* of the closure, as Table 4 does. Three facts organise the groupoid.

1. *The arrows from an object to itself are the point group.* $\mathrm{Hom}(\mathbf{G}, \mathbf{G}) = \{\mathbf{P} : \mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}\}$ is the lattice point group written in that basis (Himes & Mighell, 1987).

2. *The arrows between two objects are one coset of the point group.* If $\mathbf{P}_0$ is one arrow from $\mathbf{G}$ to $\mathbf{G}'$, every arrow between them is $g\mathbf{P}_0$ with g a symmetry of

$\mathbf{G}$, and equally $\mathbf{P}_0 g'$ with g' a symmetry of $\mathbf{G}'$ (Flack, 1987).

3. *The closure is finite and complete.* A lattice has $n = 1, 2, 3, 6$ or 16 obtuse superbases according to the zero pattern, and $48n$ labelled ones (Kurlin, 2022; Lemma A1 gives the counts independently); every arrow is a word of flips at zero conorms followed by a swap (Lemma A3), and every arrow belongs to one fixed set $\mathcal{U}$ of 6096 integer matrices (Section 3.3).

The groupoid is connected, so it is its vertex group, the point group, together with the record of the settings; what it adds to the point group is that record. The point group acts on the n unlabelled obtuse superbases. The stabiliser of a superbase F_0 is the set of symmetries that merely relabel it, its *visible* symmetries, and its orbit is the set of superbases of the same *shape*, those whose conorm vectors agree with F_0's up to a swap: a symmetry preserves $\mathbf{G}$ and hence the conorms, and conversely the relabelling that matches two equal conorm vectors is an arrow $\mathbf{P}$ with $\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}$, a symmetry. By the orbit–stabiliser theorem the order of the point group is the number of visible symmetries times the number of superbases of F_0's shape, and the number of shapes is the number of orbits. All its superbases carry the same root invariant — the six conorm values arranged in their three pairs of opposite slots, ab with cd, ac with bd, ad with bc — which Kurlin (2022) proved to be a complete isometry invariant of the lattice; the invariant identifies the lattice, and the groupoid records how its settings are related. A reduced cell here is always an obtuse superbase, not a Niggli cell; a Niggli cell of type II is an obtuse superbase, a Niggli cell of type I is not.

3.2. One monoclinic cell

Take $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$. Its reduced superbase is $M_0 = (d, a, b, c)$ with conorms

$$p_{da} = 639.5, \quad p_{db} = 1600.0, \quad p_{dc} = 2239.5, \quad p_{ab} = 0, \quad p_{ac} = 260.5, \quad p_{bc} = 0 \text{ Å}^2.$$

The two zeros are the right angles α and γ at the twofold axis b, present in every primitive monoclinic lattice.[z46] The closure has six superbases (Table 3); each has two zero conorms, each flip leads to a neighbour, and the six form a ring M_0–M_1–M_5–M_4–M_3–M_2–M_0 (Fig. 1). They fall into three shapes: $\{M_0, M_4\}$ is 30, 40, 50 Å, $\beta = 100°$; $\{M_1, M_3\}$ is 30, 40, 53.7 Å, $\beta = 113.4°$, the alternative choice $a{+}c$; $\{M_2, M_5\}$ is the third cell choice, on c and $a + c$.[z46]

Table 3. *The Selling closure of the monoclinic cell $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$: six obtuse superbases, each up to the order and common sign of its vectors, the flips at zero conorms that connect them, and where the four symmetries of the point group $2/m$ carry M_0.*[z46]

object	four vectors	flips lead to	symmetries carry M_0 here
M_0	$-a-b-c,\ a,\ b,\ c$	$M_1,\ M_2$	1, $\bar{1}$ (swaps of M_0)
M_1	$-b-c,\ -a,\ b,\ a+c$	$M_0,\ M_5$	
M_2	$-a-c,\ a+b,\ -b,\ c$	$M_0,\ M_3$	
M_3	$a+c,\ b-c,\ -b,\ -a$	$M_2,\ M_4$	
M_4	$a-b+c,\ -c,\ b,\ -a$	$M_3,\ M_5$	2 along b, m normal to b
M_5	$a-b,\ c,\ b,\ -a-c$	$M_1,\ M_4$	

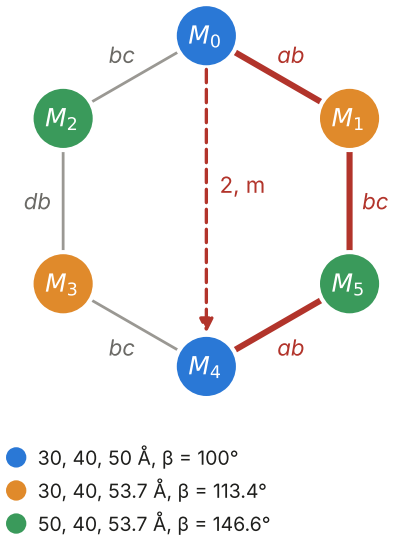


Fig. 1. The closure of the 100° cell: six superbases on a ring, joined by flips at the two zero conorms (edge labels name the slot in the labelling of Table 3) and coloured by cell shape. The twofold and the mirror carry the vectors of M_0 to those of M_4 along three flips, the word *ab.bc.db* when each flip is read in the labelling current at that step, followed by a swap; they relabel no superbase on the way, so they are invisible in the reduced cell.

The point group is $2/m$, and the last column of Table 3 shows how its four operations act on the six superbases.[z46] The identity and the inversion are swaps of M_0: they relabel it. The twofold sends $(a, b, c) \mapsto (-a, b, -c)$ and so carries the four vectors $(-a-b-c, a, b, c)$ to $(a-b+c, -a, b, -c)$, which are the vectors of M_4, three flips away; the mirror does the same. As an arrow of the groupoid each symmetry is a loop at the metric tensor of M_0, but as a word in flips and swaps the twofold is the flips ab, bc, db — the path $M_0 \to M_1 \to M_5 \to M_4$ — followed by the swap $-(a, c, b, d)$ that reads M_4 in the order in which its metric tensor is that of M_0; the mirror is the same flips with the swap (a, c, b, d).[z46] Half of the point group is therefore invisible in the reduced cell: the stabiliser of M_0 has two elements, and the orbit of M_0 is $\{M_0, M_4\}$, the two superbases of its shape, as the orbit–stabiliser count $4 = 2 \times 2$ requires. "Visible" means only that an operation relabels the reduced superbase; a search over lattice directions, as Le Page's, finds the twofold either way.

3.3. The universal operator set

The arrows can be characterised with no lattice at all. An integer matrix $\mathbf{W}$ acts on the six conorms by a fixed integer 6×6 matrix, and $\mathbf{W}$ is an arrow of the closure of every lattice with zero pattern Z exactly when that matrix maps the face $\{p \geq 0,\ p_Z = 0\}$ into the cone of non-negative conorms, a sign condition enumerated exactly (supporting information, Section S3). The union over the 38 faces is a set $\mathcal{U}$ of $6096 = 48 \times 127$ integer matrices with entries in $\{-3, \ldots, 3\}$, closed under inversion, containing every arrow between two reduced cells of any lattice.[z52] Up to swaps it has 127 elements; the superbases $\mathcal{U}F_0$ are every superbase that can be obtuse together with F_0 for some metric. This set carries the completeness statements of Section 4.

4. Measured cells: the widened groupoid

A measured cell is never exactly on a zero conorm: a right angle is measured as 90.25° and the conorm is a small number of either sign. The flips that joined the superbases of Table 3 now lead to superbases with a slightly negative conorm, Selling reduction rejects them, and the exact closure of the measured metric has one superbase instead of six; the twofold and the mirror, which were arrows to M_4, now lead to a cell outside the set. The closure has to be widened.

4.1. The strain norm

The measure used is the relative change of the metric tensor,

$$\|\Delta\mathbf{G}\|_{\mathbf{G}} = \|\mathbf{G}^{-1/2}\,\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F, \tag{2}$$

a strain: the norm of the affine-invariant Riemannian metric on positive definite matrices at $\mathbf{G}$ (Pennec *et al.*, 2006), which is the geodesic distance to first order in $\Delta\mathbf{G}$. It is unchanged by a change of basis, since $\mathbf{G} \mapsto \mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P}$ and $\Delta\mathbf{G} \mapsto \mathbf{P}^{\mathsf{T}}\Delta\mathbf{G}\mathbf{P}$ leave it as it is, so it is the same in every frame, every arrow of the closure is an isometry for it, and it is unchanged by a common scale of the cell. A perturbation of norm ϵ changes the dot product of any two lattice vectors by at most $\epsilon|u||v|$ (Proposition A4), so the cosine of any angle by at most ϵ and a squared edge by at most a fraction ϵ, whatever the shape of the cell. An absolute tolerance on the conorms in Å^2, as is implicit in a tolerant reduction, is the wrong measure for a cell whose edges differ in length: for a needle the error on the conorm between the two long vectors is hundreds of times that between the short ones, and the frame in which a conorm is read changes its tolerance. Appendix B gives the theory for that simpler band, which is the special case of the strain norm for cells of similar edge lengths.

The classical tolerances become one kind of number in this norm. Changing the angle θ between two vectors by δ is a strain of $\delta\sqrt{2(1+\cos^2\theta)}/\sin\theta$ to first order,

$\sqrt{2}\sin\delta$ exactly for an orthogonal pair; changing one edge by a fraction f is a strain of about $2f$. Le Page's search (Le Page, 1982) accepts a twofold whose obliquity δ, the angle between a lattice row and the reciprocal row nearest to it, is at most 3°; for an axis of an orthogonal cell the obliquity is the angle by which that axis misses the normal to the other two, and the strain distance from the cell to the nearest lattice in which the axis is an exact twofold is $\sqrt{2}\sin\delta$, which is 0.074 at 3°. That number is a distance to a sort; the price of the twofold itself is twice it (Proposition A9), a distinction that Section 4.5 returns to. A 5% edge tolerance is a strain of 0.10. The norm weighs all directions of the metric alike; with a covariance for the metric tensor a Mahalanobis form replaces it, which stays basis-invariant when the covariance transforms with the metric. That extension is not pursued here.

4.2. The widened set

The *widened closure* of a metric $\mathbf{G}$ at band ϵ is

$$\mathcal{S}_\epsilon(\mathbf{G}) = \left\{ F \;:\; F \text{ is obtuse for some metric within } \epsilon \text{ of } \mathbf{G} \text{ in the norm (2)} \right\}. \quad (3)$$

Its superbases are the reduced cells of every lattice consistent with the measurement at band ϵ, and its arrows the unimodular matrices between their labellings. Each superbase has a *depth*, the strain distance from the measured metric to the nearest metric that makes it obtuse, a non-negative least-squares problem in its six conorms: a superbase belongs to $\mathcal{S}_\epsilon(\mathbf{G})$ when its depth is at most ϵ, and its depth is the band at which it enters. Each arrow $\mathbf{P}$ from $\mathbf{G}_1$ to $\mathbf{G}_2$ has a *price*, the strain $\|\mathbf{P}^{\mathsf{T}}\mathbf{G}_1\mathbf{P}-\mathbf{G}_2\|_{\mathbf{G}_2}$ by which it fails to carry one metric onto the other, zero for every arrow of an exact closure. Equation (3) names no starting cell, order of operations or frame, so $\mathcal{S}_\epsilon(\mathbf{G})$ is the same whoever computes it. Because the norm (2) is the linearisation of the affine-invariant distance at one metric, prices measured at different metrics agree only to first order: a price of ϵ measured at $\mathbf{G}_2$ is at most $\epsilon/(1-\epsilon)$ measured at $\mathbf{G}_1$ (Corollary A8),

and the price of an inverse arrow equals that of the arrow to first order. Within that approximation the arrows of price at most ϵ from the measured cell to itself, written H_ϵ, take the place of the point group. They need not form a group, since the price of a product is bounded only by the sum of the prices; Proposition A10 gives a condition under which they do, and the implementation checks closure directly. Two properties make $\mathcal{S}_\epsilon(\mathbf{G})$ computable and useful.

It is finite. Every vector of a superbase in $\mathcal{S}_\epsilon(\mathbf{G})$ lies in a ball of squared radius $\frac{1+\epsilon}{1-\epsilon}\mu_c$ about the origin, with μ_c the squared length of the shortest lattice vector in its parity class (Lemma A6); the lattice vectors in these seven balls are finitely many, so $\mathcal{S}_\epsilon(\mathbf{G})$ is enumerated exactly, with no window of coordinates to assume.

Widening one side suffices. If a new cell, carried to a reference by an operator $\mathbf{P}$, differs from it by a price of at most ϵ measured at the new cell, then the new cell's reduced superbase, carried by $\mathbf{P}$, lies in the reference's widened closure at band $\epsilon/(1-\epsilon)$ (Corollary A8). So the reference alone is widened, by that slightly larger band, and the new cell is reduced once.

4.3. Computing the widened set

For a measured cell and a band $\epsilon < 1$:

1. Selling-reduce the measured cell to a superbase F_0.
2. Enumerate the lattice vectors in the seven balls of Lemma A6 and form the superbases they make.
3. Keep the superbases of depth at most ϵ. Record each one's depth and the operator $\mathbf{W}$ from F_0 to it.

This is exact for every cell and every band below 1, and it is what the implementation does; the balls are small at the bands in use, and at a band of 0.08 the enumeration

takes well under a second for a cell of ordinary shape and a few seconds for a needle or plate in this implementation. Two further descriptions of the set are useful, one for understanding how far it reaches and one for relating it to the simpler constructions of Appendix B.

Layers. Call two superbases *adjacent* when some metric makes both obtuse; the superbases adjacent to F_0 are the first layer $\mathcal{N}_1 = \mathcal{U}F_0$, 127 superbases up to relabelling, and $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$, with 2755 in the second layer.[z59] The *adjacency radius* $\delta_k(\mathbf{G})$ is the strain distance from $\mathbf{G}$ to the nearest metric for which a superbase outside $\mathcal{N}_k$ is obtuse. If $\epsilon < \min(\delta_k, \frac{1}{4})$, the widened set lies in the kth layer and is obtained by filtering it, keeping the superbases of $\mathcal{N}_k$ of depth at most ϵ (Theorem A7). The *layer* of a cell at band ϵ is the least k for which $\mathcal{S}_\epsilon(\mathbf{G}) \subseteq \mathcal{N}_k$; the implementation reports it. Computing δ_k exactly costs the same ball enumeration as computing the set, at $\epsilon = \frac{1}{4}$, so the layers are not a cheaper route to a certified answer: they describe the answer. Filtering the first layer alone, 127 depths, is the cheap uncertified route, exact whenever $\epsilon < \delta_1$; Section 5.3 reports δ_1 for a range of cells, above a quarter strain for the cells of ordinary shape tried and falling with the ratio of shortest to longest edge for needles and plates. No closed form for δ_1 is proved here.

Projection. The set can also be built from the lattices near the measurement: for each of the 127 frames of the first layer and each of the 38 faces of the cone, project the measured metric onto the face (the nearest metric in the norm (2) with those conorms zero and the others non-negative); for every projection within ϵ, enumerate the exact closure of the projected lattice; keep the members of depth at most ϵ; take the union. Below δ_1 this returns $\mathcal{S}_\epsilon(\mathbf{G})$ (Theorem A7); the checks use projections within 4ϵ, a margin the proof does not need. Setting the small conorms of F_0 to zero, the construction of Appendix B, is the single projection onto one face in one frame, and a single projection is not enough (Section 5.3).

Ties take no band. Only zeros enter the construction. Every sort lies on one zero pattern, and every symmetry of every lattice on that pattern is one of its arrows (Lemma A3), so a near-equality — the $a \approx b$ of a pseudo-tetragonal cell — never adds a superbase: the operator that would be its symmetry is already enumerated, and whether it is accepted is decided by its price.

4.4. Distance to each sort

The same norm measures how far a measured cell is from each of the 24 sorts: project its conorms, in each of the 6096 frames of $\mathcal{U}$, onto the sort's subspace of Proposition A5 (its zeros and equalities, the other conorms non-negative), and take the smallest distance. Projection onto Bravais-lattice subspaces is the approach of Andrews & Bernstein (2014) in G^6 and of Oishi-Tomiyasu (2012) in Selling coordinates; here the frames are the complete finite set of the closure and the norm is the same in every frame. We call the 24 distances the *sort spectrum* of the cell. A cell idealised onto one sort can be projected again onto the next; the step between two sorts joined by a specialisation (Table S2) is the distance from the idealised upper sort to the lower one.

4.5. Reindexing

A reference R is stored as its widened closure, the operators $\mathbf{W}_k$ from its reduced cell to the members. A new cell N is reduced once, to a basis B_N in its own coordinates; B_R is the reduction matrix of the reference. For each stored $\mathbf{W}_k$ and each of the 48 swaps $\mathbf{Q}$ the candidate operator from the reference's input basis to the new cell's is

$$\mathbf{P} = B_R \mathbf{W}_k \mathbf{Q}\, B_N^{-1}, \tag{4}$$

so that $(a_N\, b_N\, c_N) = (a_R\, b_R\, c_R)\mathbf{P}$, $\mathbf{G}_N \approx \mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P}$ and Miller indices transform as $h_N = \mathbf{P}^{\mathsf{T}} h_R$. Each candidate is priced by $\|s\mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P} - \mathbf{G}_N\|_{\mathbf{G}_N}$, with s the common

scale that minimises that norm, since a detector-distance or wavelength error scales every conorm alike and the widened set itself does not depend on the scale; candidates of price at most ϵ are returned. For the true operator to be among them whenever its price is at most ϵ, the reference is widened at $\epsilon/(1-\epsilon)$ (Corollary A8), 0.080 for $\epsilon = 0.074$. A centred reference is first taken to a primitive cell and the centring matrix composed with the operator found, which is then rational in the conventional basis; the result is carried to the conventional setting of the target space group outside the closure. For a chiral crystal the candidates of determinant -1 are dropped.

Cosets and classes. Let H_ϵ be the arrows of price at most ϵ that carry the reference onto itself, and p_0 the price of the true operator $\mathbf{P}_0$. To first order in the prices every candidate is $h\mathbf{P}_0$ with h a self-map of price at most $\epsilon + p_0$, and every h of price at most $\epsilon - p_0$ gives a candidate; if no self-map of the reference has a price in $(\epsilon - p_0, 2\epsilon]$, then H_ϵ is a group and the candidates are exactly the coset $H_\epsilon\mathbf{P}_0$ (Proposition A10). The implementation checks closure of H_ϵ directly. Two candidates that differ on the left by an element of the reference's Laue group $\mathcal{L}$ index the crystal the same way, so the distinct indexings are the classes of $\mathcal{L}$ among the candidates, $[H_\epsilon : \mathcal{L}]$ of them when the coset statement holds; for anomalous data the point group replaces the Laue group, since the inversion then changes the indexing. Membership of the true operator is what is guaranteed; the price orders the candidates but does not single out the true one within its class, which is a matter for the intensities, as in a twinning test.

Which pseudo-symmetries are offered. A twofold g of a nearby lattice enters H_ϵ when its price is at most ϵ, and its price is twice the distance from the measured metric to the subspace of metrics that g leaves invariant (Proposition A9), which contains the subspace of the sort that g belongs to. A band of $\epsilon = 0.074$ on prices therefore offers the pseudo-twofolds whose obliquity is at most about 1.5°; to offer every twofold that a Le Page search at 3° would accept, the band is 0.148, still within the range in which

the computation is proved exact. The distinction is between a tolerance on how much two measurements of one crystal may differ, which is what ϵ is, and a tolerance on how far a lattice may be from a symmetry it is credited with; the two are related by the factor of two. The procedure is implemented as `agentsg.cell.strain_closure`, with `widened_closure`, `reindex`, `indexing_classes`, `price_spectrum` and `sort_spectrum`.

4.6. Verification

Exact closures and the closures of the 24 sorts were checked against brute force over every basis with coordinates in $[-2, 2]$ relative to F_0. That window contains every member of every exact closure: Lemma A1 confines the members to superbase coordinates of modulus at most 7, and the exhaustive enumeration of that larger window finds members with coordinates in $\{-1, 0, 1\}$ only.[z52,z56] Widened sets were checked against the ball enumeration of Lemma A6, which needs no window, and against brute force over $[-3, 3]$ for needles and plates.[z58,z59] Test lattices were specified by their conorms. The measured worked cell is 30.02, 39.97, 50.03 Å, 90.25°, 92.05°, 89.80°: the 100° cell of Section 3.2 with β moved to 92°, so that it sits near two zero conorms and not far from a third, perturbed by a metric error of strain 0.0082 (14.4 Å^2 in Frobenius norm). A benchmark of 780 measured cells with exact references and true operators, built where the closure predicts trouble, is described in the supporting information (Section S2). The checks are the test files listed in the supporting information (Section S4); each mark [*znn*] is placed at the end of the sentence it supports.

5. Results

5.1. The closures of the 24 sorts

For each zero pattern Z the arrows are the operators of the universal set that map the face of Z into the cone; on the face every arrow permutes the six conorm slots, so

each fixes a subspace of conorm vectors, and the sorts are the closed sets of equalities obtained by intersecting those subspaces (Proposition A5). For a sort, the point group is the set of arrows that fix the sort's subspace pointwise, the visible symmetries are those that are swaps of the reduced cell, and two superbases have the same shape when their conorms agree up to a slot permutation. Every entry of Table 4 is the result of this linear algebra on integer matrices.[z54] Out of each labelled superbase there are exactly as many symmetries as the point group has elements, and in every sort that number is the order of the holohedry: 48, 24, 16, 12, 8, 4 or 2. The twofolds, with the inversion, generate the group in every sort, so finding the twofolds suffices.[z46]

Table 4. *The Selling closure of each Delone sort, computed from the operators alone. Patterns are Selling parameters in the* S^6 *order* $b\cdot c, a\cdot c, a\cdot b, a\cdot d, b\cdot d, c\cdot d$*; equal letters denote equal values and* 0 *a zero. Superbases are counted up to swaps. "Visible" is the number of symmetries that relabel the reduced superbase; the rest carry it to another superbase. "Shapes" is the number of distinct cell shapes among the superbases, equal in every case to the number of orbits of superbases under the symmetries. "Visible" depends on which superbase one reduces to; values are for the pattern as written.*[z54]

sort	Bravais	pattern	zeros	superbases	|group|	visible	shapes
C1	cI	rrrrrr	0	1	48	48	1
C3	cF	rr0rr0	2	3	48	16	1
C5	cP	000rrr	3	16	48	12	2
H4	hP	00rrrs	2	6	24	4	1
T1	tI	rrsrrs	0	1	16	16	1
T2	tI	rr0rrs	1	2	16	8	1
T5	tP	000rrs	3	16	16	4	3
R1	hR	rrrsss	0	1	12	12	1
R3	hR	rr0sr0	2	3	12	4	1
O1A	oF	rrsrrt	0	1	8	8	1
O1B	oI	rstrst	0	1	8	8	1
O2	oI	rs0srt	1	2	8	4	1
O3	oI	rs0rs0	2	3	8	8	2
O4	oS	00rsst	2	6	8	4	2
O5	oP	000rst	3	16	8	2	4
M1A	mC	rrsttu	0	1	4	4	1
M1B	mC	rstrsu	0	1	4	4	1
M2A	mC	rs0stu	1	2	4	2	1
M2B	mC	rs0rst	1	2	4	4	2
M3	mC	rs0ts0	2	3	4	4	2
M4	mP	00rstu	2	6	4	2	3
A1	aP	rstuvw	0	1	2	2	1
A2	aP	rs0tuv	1	2	2	2	2
A3	aP	rs0tu0	2	3	2	2	3

Zero pattern and sort fix different parts. Which superbases exist and which arrows

join them is fixed by the zero pattern; which arrows are symmetries is fixed by the sort, and the abstract group by the Bravais type. Primitive monoclinic, C-centred orthorhombic and primitive hexagonal lattices all have two zeros sharing a vector and six superbases, and differ only in how many arrows are symmetries (4, 8, 24). Within one sort the shape of the cell plays no part: the superbases, arrows and symmetries are the same, and the lattice supplies only the numbers attached to them.

Symmetry visible in one cell. In the sorts with zero conorms the symmetries that relabel the reduced cell are a minority: 2 of 8 for primitive orthorhombic, 4 of 16 for primitive tetragonal, 12 of 48 for primitive cubic, 4 of 24 for hexagonal, 4 of 12 for rhombohedral R3.[z46] None of the three twofolds of a primitive orthorhombic lattice relabels its reduced cell, nor the one twofold of primitive monoclinic; of the five, nine and seven twofolds of primitive tetragonal, cubic and hexagonal lattices, one, three and one do.[z46]

Cell shapes. The columns of Table 4 obey the orbit–stabiliser count of Section 3.1: the orbit of the reduced cell under the point group has |group|/visible superbases, all of its shape, and the number of shapes is the number of orbits, computed independently in the check.[z54] For cP the orbit of the orthogonal cell has $48/12 = 4$ superbases, the four orthogonal superbases, and the twelve others form the second shape; for tP it has $16/4 = 4$, for oP $8/2 = 4$, and for hP all $24/4 = 6$ superbases are one shape. No symmetry relates cells of different shape. Two measurements of the monoclinic lattice that reduce to M_0 and to M_1 report different cell parameters and are related by no symmetry, only by the flip at *ab*. Ten of the 24 sorts have more than one shape. Primitive orthorhombic has four: the orthogonal cell and three monoclinic-looking cells with two right angles, one for each axis traded for a face diagonal; reduction of a measured orthorhombic cell stops at whichever of the sixteen superbases its errors favour, and the orthogonal cell is up to three flips away, reached by no twofold of the

lattice. The three shapes of a primitive monoclinic lattice are the three cell choices of *International Tables* (Hahn, 2006, Section 2.2.16); the catalogue is Table S1 of the supporting information.

Specialisations. The same algebra gives the conditions under which a lattice of one sort becomes a lattice of another (Table S2 of the supporting information). Every condition of codimension one is a conorm vanishing or two conorms becoming equal; they join 36 pairs of sorts.[z50] The classical values $c/a = \sqrt{2}$ (tI to cF), $b = \sqrt{3}\,a$ (oS to hP) and $a = c$ in a monoclinic cell are among them, and the monoclinic conditions come in threes, one per cell choice. The two kinds behave differently under error. A near-equality, such as $c/a \approx \sqrt{2}$, changes no zero pattern: the set of superbases is already complete and the pseudo-symmetry appears as a priced arrow. A near-zero, such as a C-centred monoclinic cell one zero from primitive tetragonal, adds superbases, and those are what the widened closure recovers.

5.2. Distances to the sorts

The sort spectrum orders the possible symmetry assignments of a measured cell in one unit. An oP cell $40.0 \times 40.3 \times 80\,\text{Å}$ is 1.06% from tP, the strain of its 0.75% edge mismatch; a C-centred monoclinic cell $50 \times 86.8 \times 40\,\text{Å}$, $\beta = 90.4°$, is 1.04% from hP; an F cell $100 \times 100.2 \times 99.9\,\text{Å}$ is 0.43% from cF.[z58] Two properties follow from the geometry. Along every specialisation of Table S2 the distance cannot decrease, since the lower sort's subspace lies inside the upper one's;[z58] and when the two projections are made in one frame and one norm, $d^2_{\text{lower}} \geq d^2_{\text{upper}} + s^2$ with s the step from the idealised upper sort, with equality when the non-negativity constraints are inactive (Pythagoras for nested subspaces); the cells of Fig. 2 are close to equality along the cheapest route. A distance is minimised over the 6096 frames of $\mathcal{U}$, which contain the frame of the nearest lattice of the sort whenever that distance is below the adjacency

radius δ_1 (Theorem A7). A value above δ_1 is only an upper bound on the true distance, since the minimum was taken over a subset of frames; what can be asserted is that the sort is further away than δ_1. For the cells reported below δ_1 reaches the quarter cap, the worked cell and lysozyme included,[z60] so a distance above a quarter strain is reported as "$> 25\%$", a lower bound on the true distance.

Figure 2 draws the spectrum on the graph of specialisations for the worked cell; the same drawing for a tetragonal lysozyme cell indexed in $P1$ is in the supporting information. For the worked cell the cheap route is A1 to A2 at 0.28%, then mP (M4) at 0.78% and oP at 5.1%, with the third triclinic sort A3 at 3.8%; every other sort is more than 25% away.[z60] For the lysozyme cell the twelve sorts from aP to tP all lie within 0.22%, and every other sort is more than 25% away.[z60] In both cells the sorts fall into a cluster and a gap. We call the *gap* of a spectrum the largest ratio between consecutive distances in the sorted list of Bravais types, aP excluded, and the type before it the *gap type*; it is a heuristic, since no error model enters it, but a band anywhere inside the gap gives the same symmetry assignment. The gap type is mP for the worked cell and tP for lysozyme. The many C-centred sorts near lysozyme are no separate pseudo-symmetry: a square net has an equally good C-centred cell on its diagonals, so every tP lattice is also nearly oS and mC. Dashed edges keep the Bravais type and change only the reduced cell (A1 to A2 to A3, or between the mC sorts); they add superbases to the closure but no symmetry.

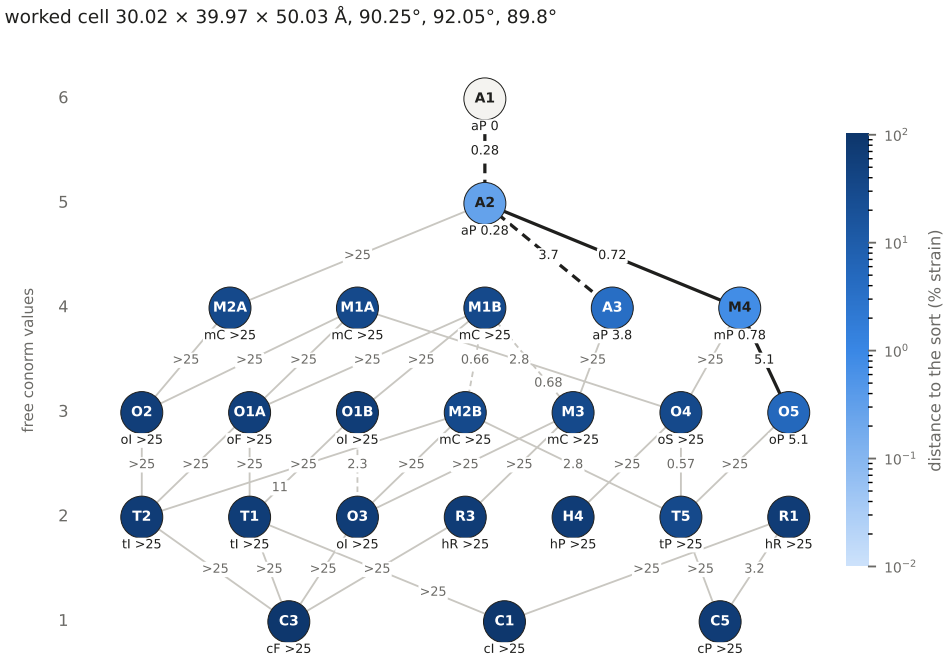


Fig. 2. The 24 Delone sorts as a graph for the measured worked cell. Rows are the number of free conorm values. Nodes give the Bravais type and the strain distance (%) from the measured cell to the sort, colour the same distance on a log scale; distances above 25% are lower bounds, since δ_1 reaches the quarter cap for this cell. Edges are the specialisations of Table S2, labelled with the step from the cell idealised onto the upper sort to the lower one. Dark edges are routes on which the steps add in quadrature to the lower sort's distance within 3%, drawn where that distance is below 10%. Dashed edges keep the Bravais type. The node distances are checked in [z60].

The seven measured cells that Oishi-Tomiyasu (2024) distributes with her program are a small test, with no ground truth beyond the program's own assignment and with the next type far away in every case, so it is a check of consistency and not of discrimination (Table 5). In every case the program's Bravais type is the last of the cluster of types within 3.6%, and the next type is at least 14.9% away, so any band between those two numbers selects it; in six of the seven the program's type is the gap type, while for the alumina cell the largest ratio in the ladder is between mC at 0.43% and hR at 3.58%. The program's distance is a norm in Å^2 on the metric tensor

(Oishi-Tomiyasu, 2012) and only its value for the assigned type is distributed, so no comparison of the full ladders is possible here.[z51,z60]

Table 5. *The seven sample cells of BLDConograph (Oishi-Tomiyasu, 2024): the program's Bravais type and its minimal distance to that type (Å^2); the Bravais types of the sort spectrum in increasing strain distance (%), aP omitted, up to and including the program's type; and the next type with its distance. For all seven cells δ_1 reaches the quarter cap, so "> 25" is a lower bound on the distance as well as the outcome of the minimum over the frames of $\mathcal{U}$.*[z51,z60]

sample	type	program	types up to the program's (strain %)	next
HERMES_Sr327_250K	tI	0.57	mC 0.31, oI 0.74, oF 0.83, tI 0.84	mP > 25
1339prbasrcuo_bs1	tP	0.056	mP 0.11, mC 0.14, oS 0.17, oP 0.18, tP 0.22	oF > 25
sample3	cI	2.07	mC 0.23, hR 0.39, oF 0.39, oI 0.49, tI 0.49, cI 0.58	cF > 25
CeO2_matsushita	cF	0.084	mC 0.09, oI 0.17, oF 0.17, tI 0.18, hR 0.26, cF 0.30	cI > 25
M402-pf-119673	hP	20.5	mC 0.99, mP 2.24, oS 2.37, hP 2.94	oI > 25
Acrinol-B-D8CuKa1	mC	0.72	mC 0.31	oF 20.6
Alumina676	hR	1.93	mC 0.43, hR 3.58	oI 14.9

5.3. Completeness on measured cells

A worked filter. The reduced superbase of the worked cell has conorms 49.5, 1588.9, 2444.7, 4.2, 847.5 and 4.5 Å^2 in the slots da, db, dc, ab, ac, bc. Its widened closure grows with the band: one member at $\epsilon = 0.001$, two at 0.003, three at 0.006, the six of the ring from 0.01, and sixteen at 0.074, the strain of one angle changed by 3°.[z60] Table 6 lists the sixteen, each as a word in the flips of Table 1, with its conorms under the measured metric and its depth (Fig. 3). The six members of the ring enter below 0.8%; the other ten members of the closure of the nearby orthorhombic lattice enter between 3.8 and 5.1%. Depths are not ordered as the most negative conorms are — da, with -49.5 Å^2, enters before $da.ab.bc$, with -45.0 — because the strain weighs each conorm by the lengths of the vectors it joins.

Table 6. *The widened closure of the measured worked cell. Each row is a member, named by a word in the flips of Table 1 from the reduced cell e; the six numbers are its conorms under the measured metric, slots da, db, dc, ab, ac, bc, in Å²; "depth" is the strain at which it enters, in %. At a band of 7.4% all sixteen are members.*[z45,z60]

word	*da*	*db*	*dc*	*ab*	*ac*	*bc*	depth
e	49.5	1588.9	2444.7	4.2	847.5	4.5	0
bc	45.0	8.7	2449.3	1593.4	852.0	−4.5	0.28
ab	851.7	1593.1	2440.6	−4.2	53.7	8.7	0.49
bc.db	2458.0	−8.7	53.7	1602.1	843.3	4.2	0.62
ab.bc	842.9	4.5	2449.3	1601.8	62.5	−8.7	0.78
ab.bc.db	2453.8	−4.5	847.5	1606.3	57.9	−4.2	0.78
da	−49.5	2494.3	1638.4	53.7	897.0	−45.0	3.76
ab.bc.ac	1664.2	−57.9	2511.7	905.4	−62.5	53.7	4.56
da.ab	950.7	2548.0	1584.7	−53.7	4.2	8.7	5.06
da.ab.ac	−49.5	2543.8	1588.9	954.9	−4.2	12.9	5.08
ab.ac	49.5	1539.3	2494.3	905.4	−53.7	62.5	5.08
da.ab.bc	942.0	−45.0	1593.4	2556.7	12.9	−8.7	5.09
bc.db.dc	45.0	2511.7	−53.7	1548.4	897.0	57.9	5.09
ab.ac.bc.ac	1610.5	959.1	2565.5	−4.2	−8.7	−53.7	5.11
da.ab.bc.ac	2569.6	−57.9	1606.3	954.9	−12.9	4.2	5.11
ab.ac.bc	−12.9	967.8	2556.7	1601.8	8.7	−62.5	5.11

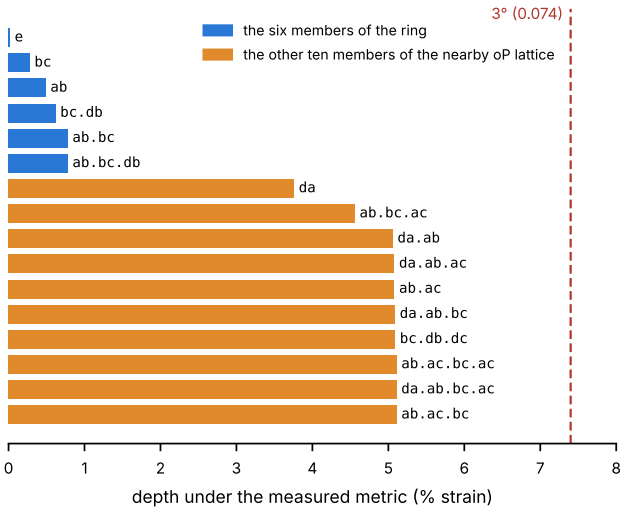


Fig. 3. The sixteen members of the widened closure of the measured worked cell (Table 6) by depth: the six of the ring enter below 0.8% strain, the other ten members of the nearby orthorhombic lattice's closure between 3.8 and 5.1%; the band selects among them.[z60]

The set has to be computed, not sampled. Reduction with a widened boundary does not stop at a well-defined cell. With a band on the conorms (Appendix B), reduction of the worked cell and of a noisy orthorhombic cell from random starting bases stopped

at several different members of the widened set, and on a noisy cubic cell the restarts did not find all of them.[z45] Walks from the measured reduced cell, flipping wherever a conorm is within a band of zero, miss members too; with the conorm band, on 80 cells with zero conorms at awkward positions, four versions of the walk returned a wrong set on 2 to 13 cells (Table S3 of the supporting information). The cause is a member reachable only through a non-member, for instance when two small conorms lie at 1.2 and 0.5 times the band, where the flip across the first leaves the band and only a second flip, across the other, returns into it.[z44,z45] Such *difference-wall members* carry no symmetry; they are ordinary alternative reduced cells to which a second measurement can reduce, and a reference set built by walking never contains them.[z46]

Needles and plates. A band on the conorms has to cover the conorms between the long vectors of a needle, and the proved conorm-band construction of Appendix B then fails at a few thousandths of a degree. Bands per slot, $\sigma|u_i||u_j|$, applied in the frame of the reduced cell, are not exact either: the needle $5.58 \times 5.21 \times 131.97$ Å, $\gamma = 91.12°$, at $\sigma = \sin 0.064°$ gives 6 superbases where the per-slot widened set has 12, because on the slot $(a+c, b)$ of the missing members the same dot product carries a much larger band.[z57] The strain norm has no frame-dependent band. The projection construction of Section 4.3 returned the exact set for this needle at $\epsilon = 0.0005$, 0.002, 0.005, 0.01, $\sin 1°$ and $\sin 3° = 0.052$ (a band set by an angle of 3° read as a cosine change, the convention of the checks, smaller than the 0.074 used for reindexing); for a $100 \times 100.1 \times 5$ Å plate and a $150 \times 120 \times 4.8$ Å plate, $\beta = 90.4°$, at $\epsilon = 0.052$, where their widened sets have 20 and 28 members, more than any single closure; and for 60 random needles, plates, isotropic and monoclinic cells at four bands each, $\sin 0.1°$ to $\sin 3°$, in all 240 cases, against brute force over coordinates in $[-2, 2]$.[z58,z60] A single projection onto the most special face is not enough: on the needle at $\epsilon = 0.002$ and

0.005 it returns 6 of the 8 members, missing those built on $a + c$, which belong to a lattice in which $(a + c) \cdot b$, not $a \cdot b$, is zero.[z60]

Adjacency radii. Table 7 gives the adjacency radii. For the worked cell and the $10 \times 10 \times 30$ Å needle no superbase outside the first layer comes within a quarter strain, so filtering the first layer is exact at every band in use; for needles and plates δ_1 falls with the ratio of the shortest to the longest edge, between one and $\sqrt{2}$ times that ratio in the four cells of the table with a finite δ_1, and the second layer roughly doubles it. The two plates at $\epsilon = 0.052$ lie between δ_1 and δ_2: the first layer misses members there, and the second contains them all, as Theorem A7 requires.[z59] Their δ_2 also exceeds 0.080, the band at which a reference is widened for $\epsilon = 0.074$, so at that band the filtered second layer is their widened set. The layer in which the set lies is a measure of a cell's difficulty, in place of a single angular limit; the exact enumeration does not depend on it.

Table 7. *Adjacency radii in the strain norm (2): δ_1, the smallest strain at which a superbase outside the first layer becomes obtuse, and δ_2 for the second layer. The search is capped at* 0.25. *A sort distance of* 0.074 *is one cell angle changed by* 3°.[z59]

cell	δ_1	δ_2
$30 \times 40 \times 50$ Å, $\beta = 92°$ (the worked cell)	> 0.25	> 0.25
needle $10 \times 10 \times 30$ Å	> 0.25	> 0.25
needle $10 \times 10 \times 100$ Å	0.139	> 0.25
needle $5.58 \times 5.21 \times 132$ Å, $\gamma = 91.12°$	0.056	0.111
plate $100 \times 100.1 \times 5$ Å	0.050	0.141
plate $150 \times 120 \times 4.8$ Å, $\beta = 90.4°$	0.043	0.100

5.4. Reindexing

Indexing classes. Table 8 and Fig. 4 give the classes for six references, each against the same crystal measured again with a 0.1% error in the length scale and a relabelling of its axes, at $\epsilon = 0.074$. For the exactly symmetric references the counts are the index of the Laue group in the lattice holohedry, which is the classical table of indexing ambiguities for merohedral crystals (Yeates, 1997; Dauter, 2003; Evans, 2006, 2011): tetragonal lysozyme in $P4_32_12$ has one class, $P3$ has four, joined by the twofolds

along [001], [110] and [1$\bar{1}$0], and $P3_121$ and $P2_13$ have two, joined by one merohedral twofold. What the closure adds is the priced classes: a pseudo-tetragonal $P2_12_12_1$ cell, $40.0 \times 40.3 \times 80$ Å, and the worked cell in $P1$ each have a second class at a price, 2.11% and 1.55%, from the twofold that would make the cell tetragonal or monoclinic. That price is twice the cell's distance to the sort the twofold fixes, 1.06% and 0.78% (Proposition A9),[z60] so a pseudo-merohedral ambiguity is a priced alternative rather than a free one, and the band decides whether it is offered: along a scan of β for a monoclinic $30 \times 40 \times 50$ Å cell the pseudo-orthorhombic class is offered at $\epsilon = 0.074$ for $\beta = 91.4°$ and not for 91.6°, and at $\epsilon = 0.148$ for 92.5° and not for 93.5°, as the rule predicts (the limits 91.5° and 93°).[z60] For a chiral crystal the improper candidates are dropped, which halves H_ϵ and leaves the class count unchanged, and anomalous data then change nothing: lysozyme keeps one class and $P3$ four. For an acentric crystal that is not chiral the inversion separates indexings when anomalous differences are used, and the point group replaces the Laue group: $P4mm$ goes from one class to two and $P31m$ from two to four.[lib]

Table 8. *Distinct indexings of six references at* $\epsilon = 0.074$ *(the reference widened at* 0.080*): the order of the band's self-map group* H_ϵ *and of the Laue group* $\mathcal{L}$*, the number of classes of* $\mathcal{L}$ *among the candidates, and the price of each class; the new cell is the reference measured again with a* 0.1% *scale error and relabelled.*[lib,z60]

reference	space group	$\lvert H_\epsilon \rvert$	$\lvert \mathcal{L} \rvert$	classes	prices (%)
$79.1 \times 79.1 \times 37.9$ Å	$P4_32_12$	16	16	1	0
$a = b = 50$, $c = 80$ Å	$P3$	24	6	4	0, 0, 0, 0
$a = b = 50$, $c = 80$ Å	$P3_121$	24	12	2	0, 0
$a = 60$ Å	$P2_13$	48	24	2	0, 0
$40.0 \times 40.3 \times 80$ Å	$P2_12_12_1$	16	8	2	0, 2.11
worked cell	$P1$	4	2	2	0, 1.55

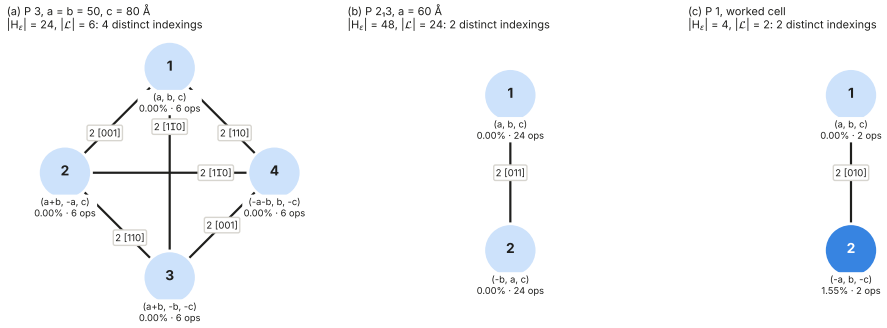


Fig. 4. Three of the cases of Table 8 as graphs: the classes of the Laue group among the candidates for (*a*) $P3$, (*b*) $P2_13$ and (*c*) the worked cell in $P1$. Each node is one class; it shows the class's operator nearest the identity (new basis in terms of the reference's a, b, c), its strain price and the number of operators in it. Each edge is the lattice symmetry or pseudo-symmetry carrying one class to the other, as rotation order and axis $[uvw]$ in the reference basis. Node colour is the price, from 0 (light) to 3%.

The benchmark. The 780 constructed cases of the benchmark (supporting information, Section S2) are primitive cells with a random metric error and a random reindexing, built where the closure predicts trouble: 75 cells measured in another shape of the same lattice, 660 lattices a small distance from a specialisation, 20 whose only reduced superbase is a difference-wall member, 16 needles and plates, and 9 cells of the boundary sorts. Table 9 scores two methods on them. The closure, with the reference widened at 0.080 and candidates accepted at $\epsilon = 0.074$, had the true operator among its candidates in all 780 cases, and in all 778 at a band of three times each case's own error strain (two cases have an error above the proved range at that multiple). The second method is the search that data-processing programs make: Selling-reduce both cells and try every unimodular matrix with entries in $\{-1, 0, 1\}$ between the two reduced bases, 6960 matrices, priced in the same norm. It found the true operator in 778 of 780, missing two triclinic cells one zero from mP whose operator between the reduced bases has an entry of 2, at a cost comparable to the closure's in this implementation. A hit counts the true operator or its negative, which differ by the inversion and

so by an element of the Laue group. The errors in the benchmark are small, a median strain of 10^{-3} against the band, so the 780 of 780 says that the implementation delivers the guarantee of Corollary A8, not that the hit rate is high where a small-matrix search fails; the gain is the guarantee, the coset structure and the operators outside the small window. Symmetry-only reindexing, which pairs the new reduced cell with the reference's reduced cell in its 48 labellings, proposes the true operator in 54% of the cases, and the exact closure without widening in 96%. The 16 needles and plates lay in the second layer in 12 cases and in the first in 4; no case was refused, where the conorm-band construction of Appendix B declined 8 of them. The same cases show where the Niggli cell is unstable: the measured cell's Niggli cell is a different cell from the reference's in 45 of 780, nearly all at the body-centred orthorhombic sort O3 and its neighbours, where two shapes of one closure are chosen by the sign of a near-zero quantity (`evaluate.py --niggli`). Centred references in conventional settings ($C2$, $I222$, $R3$ in hexagonal axes, $F23$, $I4_1$, $Fddd$), measured in a random primitive basis with a random error, are reindexed to the conventional basis with the rational true operator among the candidates in every case tried.[z60]

Table 9. *The benchmark of 780 constructed cases, by family: the number of cases in which the true operator is among the candidates, for the widened closure (reference widened at* 0.080*, candidates accepted at* $\epsilon = 0.074$*) and for the search over the 6960 unimodular matrices with entries in* $\{-1, 0, 1\}$ *between the two reduced bases, priced in the same norm; the layer in which the widened set lay; and the* 10*th,* 50*th and* 90*th percentiles of the number of candidates. Results in* `manuscript/benchmark/results_strain.txt`*.*

family	cases	closure	small matrices	layer 1/2	candidates
shapes	75	75	75	75/0	4/8/16
specialisation	660	660	658	660/0	4/8/48
difference-wall	20	20	20	20/0	8/21/48
needles and plates	16	16	16	4/12	24/60/80
boundary sorts	9	9	9	9/0	4/4/8
all	780	780	778	768/12	4/8/48

Completeness of the stored set matters. A new cell constructed so that its only reduced superbase is a difference-wall member of an orthorhombic corner was matched, with the conorm band of Appendix B, against two stored sets of the reference. The

complete set offers three candidates and their negatives; a set built by idealising only at the band has lost the third, which is $\pm\mathbf{I}$, the true operator.[z46] The cheapest candidate is found either way, through a pseudo-symmetry, but it is not the true operator: price orders the coset, it does not identify its true member.

6. Discussion

Where reindexing has been hard. The usual pipeline reduces a cell, idealises it to the highest Bravais lattice within a tolerance (Le Page, 1982; Sauter *et al.*, 2004) and standardises to a conventional cell; two data sets so treated agree unless they idealise to different lattices or fall on opposite sides of a tie in the standardisation. What the symmetry step cannot propose is the flip at ab, because it is not a lattice symmetry of any cell. That flip separates c from $a + c$, the orthogonal cell from its monoclinic-looking shapes, and the two reduced cells of the sort O3 that two measurements return. The closure adds exactly those operators, with the pseudo-symmetries, each priced, and its gain over the pipeline is the completeness statement, not a higher hit rate on ordinary cells. Table 10 collects the traps, where each is found and what resolves it.

Table 10. *The reindexing traps as the closure finds and resolves them.*

trap	mechanism	found	resolved by
monoclinic c versus $a + c$	flip at the zero conorm ab; two shapes, no symmetry between them	Sec. 3.2, Table 3	both shapes are superbases of the closure
oP, tP, cP reported as a monoclinic cell	twelve of sixteen superbases lie on face diagonals	Sec. 5.1, Table 6	the widened closure; the orthogonal cell and its twofolds enter at a depth
pseudo-merohedry	a near-equality leaves the set of superbases complete; a near-zero adds superbases	Sec. 5.1, Fig. 4	prices; indexing classes
Niggli-cell instability (O3 and neighbours)	two shapes of one closure chosen by the sign of a near-zero	Sec. 5.4	both shapes stored
difference-wall member	reachable only through a non-member	Sec. 5.3, Table S3	enumerate the set, do not walk
needles and plates	a conorm's error depends on the slot and the frame	Sec. 5.3, Table 7	strain norm; second layer
choice within the coset	the metric cannot separate an operator from its product with a (pseudo-)symmetry	Sec. 5.4	not by the metric: intensities

The Niggli cell. The instability of the Niggli cell (Andrews *et al.*, 1980; Andrews & Bernstein, 1988, 2014) is the same phenomenon only in part. Andrews & Bernstein locate it at every boundary of the Niggli cone: the sign boundary between type I and type II, the ties $a = b$ and $b = c$, and the special conditions. Only the boundaries at a zero scalar product are zero conorms; a tie is not, and the type I/II change gives a cell of the same lengths and supplementary angles, which in the closure is one shape and a swap. The Selling-reduced cell shares the instability at zero conorms, as Andrews *et al.* (2019*b*) note; what the closure adds is not stability of any one cell but completeness of the set of alternatives, so that two data sets reduced either way are related by a stored operator. Tolerant Niggli reduction (Křivý & Gruber, 1976; Grosse-Kunstleve *et al.*, 2004) returns one of these cells, not the set.

Relation to S^6 and SELLA. In the language of Andrews, Bernstein & Sauter (2019*a*) a flip is a boundary transformation of S^6 and a swap, acting on the six Selling parameters, is one of the 24 reflections of S^6 (the common sign change acts trivially on dot products), so the closure of a lattice on a face is the set of its images under those maps

that stay in the all-non-positive orthant of S^6; the 6096 matrices of $\mathcal{U}$ are 48×127 on bases where 24×127 act on S^6. SELLA (Andrews *et al.*, 2023) carries the boundary images of a cell to measure its distance to each sort in S^6, and the projection onto the subspace of a Bravais type goes back to Andrews & Bernstein (1988) in G^6. What is added here is the groupoid packaging, with the counts of visible symmetries and shapes, the universal set with its layers, and a norm that is the same in every frame; a direct comparison of the sort spectrum with SELLA or with the S^6 distance on the same cells is not made in this paper, and the one S^6 computation in the checks, on the worked pair, agrees with the closure (supporting information, z42). The twofolds are the operators that matter: their prices, twice the distance to the subspace they fix, rank the pseudo-symmetries as the tolerance-ranked subgroups of Sauter *et al.* (2004, 2006) and the twinning operators of Lebedev *et al.* (2006) do, and the indexing classes are the twin laws of the reference, as tabulated for merohedral cases by Yeates (1997) and Dauter (2003) and resolved from intensities by Evans (2006, 2011), Brehm & Diederichs (2014) and Gildea & Winter (2018). Reference-based reindexing in data-processing programs (Kabsch, 2010; Evans, 2006) searches small integer matrices between reduced cells; Section 5.4 compares against that search.

What is proved and what is checked. Table 11 separates the two. The proofs rest on Selling's identity, the Conway–Sloane characterisation of obtuse superbases and Kurlin's enumeration, whose counts are reproduced here by brute force for every sort; the computer-assisted parts are marked in the table.

Table 11. *What is proved, what is checked by computation, and what is refuted.*

statement	status
every arrow is flips at zero conorms and a swap; the closure of each zero pattern lies in a window	proved (Lemmas A1, A3)
$\mathcal{U}$ contains every arrow between reduced cells of any lattice	proved by exhaustive enumeration (supporting information, Section S3)
the 24 sorts are the closed sets of equalities	definition (Proposition A5); agreement with Delone checked
the strain norm bounds every dot product	proved (Proposition A4)
the widened set is finite and enumerated exactly	proved (Lemma A6)
the filtered layer $\mathcal{N}_k$ is the widened set below δ_k; projection is exact below δ_1	proved (Theorem A7); the radii are computed per cell
widening the reference alone suffices, at $\epsilon/(1-\epsilon)$	proved (Corollary A8)
a twofold's price is twice its distance to the metrics it fixes	proved to first order (Proposition A9)
the candidates are a coset $H_\epsilon \mathbf{P}_0$ and H_ϵ a group, in a gap of the price spectrum	proved to first order (Proposition A10)
the conorm-band construction is exact when $\lambda_1 > 32\tau$; the reindexing guarantee in that norm	proved (Appendix B)
projection is exact between δ_1 and δ_2	checked on every case tested
δ_1 between 1 and $\sqrt{2}$ times shortest/longest edge	observed on six cells
H_ϵ closed under composition at $\epsilon = 0.074$ for the references of Table 8	checked
the 6096 operators of $\mathcal{U}$, the layers of 127 and 2755, the counts of Lemma A1, Proposition A5 and all adjacency radii	computer-assisted enumeration
bands per slot applied in the frame of the reduced cell	refuted (the needle of Sec. 5.3)

Scope. "Complete" in this paper means: every unimodular operator whose price is at most the band is among the candidates, for bands below a quarter strain. Operators that the closure does not contain are outside its scope: sublattice and superlattice relations (a doubled axis from weak odd layers; determinant ± 2 or more), the obverse and reverse settings of a rhombohedral lattice, and the choice of hand; in macromolecular work these are separate steps. An error larger than the band, which the user does not know in advance, is outside it too. The distance used is the linearised affine-invariant norm, and the statements that compare prices at different metrics hold to first order; the exact geodesic distance (Pennec *et al.*, 2006) would make them exact and the overlap of Corollary A8 symmetric, at the cost of the convex least-squares form of the depth. The choice within a class is left to the intensities.

7. Conclusions

All obtuse superbases of a lattice, with the integer matrices between them, form a finite groupoid: its arrows at one object are the point group, its arrows between two objects a coset of it, and no symmetry relates cells of different shape. For a measured cell the groupoid is widened to every superbase within a strain ϵ of obtuse, in a norm that is the same in every basis; the widened set is finite, is enumerated exactly, and lies in the first layer of 127 superbases for the cells of ordinary shape tried and in the second layer of 2755 for needles and plates. In a pipeline this means: store each reference once as its widened closure, at a band slightly above the tolerance ϵ by which two measurements of the crystal may differ, 0.080 for $\epsilon = 0.074$; Selling-reduce each new data set once and pair it with the stored members; group the candidates by the reference's Laue group, or its point group for anomalous data. Every unimodular operator of price at most ϵ is then among the candidates, each class carries a strain price, and a pseudo-symmetry is offered when its distance to the lattice it would fix is at most $\epsilon/2$. A search over small integer matrices between the reduced cells finds the true operator nearly as often on constructed cases; what the closure adds is the guarantee, the coset structure of the candidates, and the operators that are not lattice symmetries, which a symmetry search cannot propose.

Appendix A
Proofs

Terms. Write a lattice vector in any basis, $x = n_1 v_1 + n_2 v_2 + n_3 v_3$ with integer n_i, and record whether each coordinate is even or odd. This sorts the lattice into eight *parity classes*, the classes of $\Lambda/2\Lambda$; two vectors are in the same class exactly when their difference is twice a lattice vector, so the classes do not depend on the basis.

One class is the doubled vectors; the other seven are used here. A *partial sum* of a superbase is the sum of some of its four vectors; there are seven up to sign, one in each of the seven classes. A superbase is obtuse exactly when each of its partial sums is a shortest vector of its parity class, the Voronoi vectors of the lattice (Voronoi, 1908; Conway & Sloane, 1992; Kurlin, 2022, Lemmas 2.4, 2.7 and A.1). The *excess* $e_c(F) = |v_c(F)|^2 - \mu_c$ of a superbase F in class c is the squared length of its partial sum in that class minus the squared length μ_c of the shortest vector of the class. A partial sum is written $w = \sum s_i v_i$ with $s \in \{0,1\}^4$; a vector of the same class has coordinates $s + 2k$ with k integer. Every lattice vector $x = \sum c_i v_i$ satisfies Selling's identity

$$|x|^2 = \sum_{i<j} p_{ij}\,(c_i - c_j)^2, \tag{5}$$

whatever the signs of the conorms, because the four vectors sum to zero.

Lemma A1 (window). Let F_0 be obtuse for a lattice whose non-zero conorms connect the four vectors, and F any obtuse superbase of the same lattice. Then every vector of F lies in the parity class of a partial sum of F_0 and has superbase coordinates $s_i + 2k_i$ relative to F_0, $s \in \{0,1\}^4$, $k_0 = 0$ and $|k_i| \le 3$. *Proof.* Every partial sum of F is a shortest vector of its parity class, as is the partial sum of F_0 in that class, so their excess vanishes; by Lemma A2 the excess is $4\sum p^0_{ij}\gamma_{ij}$ with every term non-negative, so $\gamma_{ij} = 0$ on every non-zero slot, i.e. $|k_i - k_j| \le 1$ along a connected graph on four vertices, and with $k_0 = 0$ every $|k_i| \le 3$. □ Enumerating the superbases of such vectors whose six pairwise products are non-positive on the whole face gives 1, 2, 3, 6 and 16 superbases for the five kinds of zero pattern, all with basis entries in $\{-1, 0, 1\}$, independently of Kurlin's lists.[z56]

Lemma A2 (class excess identity). Let F be any superbase, $w = \sum s_i v_i$ a partial sum and w' a vector of its parity class with coordinates $s + 2k$. Then $|w|^2 - |w'|^2 = -4\sum p_{ij}\gamma_{ij}$, with $\gamma_{ij} = \Delta k(\Delta k + d)$, $\Delta k = k_i - k_j$, $d = s_i - s_j$; each γ_{ij} is a non-negative

integer, zero exactly when $\Delta k \in \{0, -d\}$. *Proof.* Equation (5), $(d + 2\Delta k)^2 - d^2 = 4\Delta k(\Delta k + d)$, and $t(t + d) \geq 0$ for integers t and $d \in \{-1, 0, 1\}$. □

Lemma A3 (generation). Every arrow of the closure, and every symmetry, is a sequence of flips at zero conorms followed by a swap. *Proof.* The superbases are reached from F_0 by flips at zero conorms (Kurlin, 2022, Lemmas 4.1–4.5; verified here[z52]), so flips and swaps connect all superbases in all labellings. A symmetry maps F_0 to an obtuse superbase of the lattice; as a matrix it is the flip path from F_0 to that superbase followed by a swap. □

Proposition A4 (the strain norm bounds every dot product). If $\|\mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F \leq \epsilon$, then $|u^{\mathsf{T}}\Delta\mathbf{G}\,v| \leq \epsilon\,|u|\,|v|$ for all vectors u, v, lengths in $\mathbf{G}$; in particular every conorm of every superbase changes by at most $\epsilon|u_i||u_j|$. *Proof.* With $X = \mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}$, $u^{\mathsf{T}}\Delta\mathbf{G}\,v = (\mathbf{G}^{1/2}u)^{\mathsf{T}}X(\mathbf{G}^{1/2}v)$, at most $\|X\|_2|u||v| \leq \|X\|_F|u||v|$. □

Proposition A5 (sorts as closed sets of equalities). Fix a zero pattern Z, let O_Z be the arrows of its closure and, for $\mathbf{W} \in O_Z$, let $\mathbf{A_W}$ be its linear action on conorm vectors (supporting information, Section S3), which on the face $\{p \geq 0,\ p_Z = 0\}$ is a permutation of the slots. For a subspace V of the face put $\mathrm{Sym}(V) = \{\mathbf{W} \in O_Z : \mathbf{A_W} = \mathrm{id}\text{ on }V\}$ and $\mathrm{Fix}(H) = \{p : \mathbf{A_W}p = p\ \forall\,\mathbf{W} \in H\}$. A lattice with conorm vector p on the face has point group $\mathrm{Sym}(\{p\})$, every symmetry being in O_Z by Lemma A3. We take as definition that a sort is a maximal set of lattices with the same kind of closure and the same symmetry operators; then the sorts of kind Z are the generic points of the closed subspaces $V = \mathrm{Fix}(\mathrm{Sym}(V))$ that meet the interior of the face, modulo the arrows that map the face onto itself. The construction is the classical one, the fixed subspace of a point group in the space of forms (Delone *et al.*, 1975), written in conorm coordinates; SELLA (Andrews *et al.*, 2023) uses the same subspaces. *Proof.* Sym and Fix are a Galois connection, so the closed subspaces are the intersections of the finitely many subspaces $\mathrm{Fix}(\mathbf{W})$; since $\mathbf{A_W}$ permutes the slots on the face, each

is a set of equalities among conorms. Enumerating the intersections gives 8, 5, 3, 5 and 3 closed subspaces for no zero, one zero, two sharing a vector, two disjoint and three, with symmetry orders $2,4,4,8,8,12,16,48$; $2,4,4,8,16$; $4,8,24$; $2,4,8,12,48$; $8,16,48$; these are Delone's 24 sorts, matched pattern by pattern with Table 4. □[z53]

Lemma A6 (the widened set is finite). If F is within $\epsilon < 1$ of obtuse for $\mathbf{G}$ in the norm (2), every vector u of F satisfies $|u|^2_{\mathbf{G}} \le \frac{1+\epsilon}{1-\epsilon}\,\mu_c(\mathbf{G})$, with μ_c the squared length of F_0's partial sum in the parity class of u. *Proof.* Let $\mathbf{G}'$ within ϵ make F obtuse. By Proposition A4 with $u = v$, $(1-\epsilon)|u|^2_{\mathbf{G}} \le |u|^2_{\mathbf{G}'} \le (1+\epsilon)|u|^2_{\mathbf{G}}$ for every lattice vector. The vectors of F are shortest in their classes for $\mathbf{G}'$, and those of F_0 for $\mathbf{G}$, so $|u|^2_{\mathbf{G}'} \le |v^0_c|^2_{\mathbf{G}'} \le (1+\epsilon)\mu_c(\mathbf{G})$ and $|u|^2_{\mathbf{G}} \le |u|^2_{\mathbf{G}'}/(1-\epsilon)$. □ The lattice vectors in these seven balls are finitely many and computable from $\mathbf{G}$, so $\mathcal{S}_\epsilon(\mathbf{G})$ is enumerated exactly.[z59]

Theorem A7 (completeness up to the adjacency radius). Two superbases are *adjacent* when some positive definite metric makes both obtuse; a superbase adjacent to F_0 is $\mathbf{W}F_0$ with $\mathbf{W} \in \mathcal{U}$ (Section 3.3), since both are superbases of that metric's closure. Let $\mathcal{N}_1 = \mathcal{U}F_0$, $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$ (127 and 2755 superbases up to relabelling for $k = 1, 2$), and let $\delta_k(\mathbf{G})$ be the smallest distance in the norm (2) from $\mathbf{G}$ to a metric for which a superbase outside $\mathcal{N}_k$ is obtuse. If $\epsilon < \min(\delta_k(\mathbf{G}), \frac{1}{4})$, then $\mathcal{S}_\epsilon(\mathbf{G})$ is the set of members of $\mathcal{N}_k$ within ϵ of obtuse, and for $k = 1$ the projection construction of Section 4.3 returns it. *Proof.* A superbase in $\mathcal{S}_\epsilon(\mathbf{G})$ is obtuse for a metric within $\epsilon < \delta_k$ of $\mathbf{G}$, so it lies in $\mathcal{N}_k$; the converse is the definition. For $k = 1$, $F = \mathbf{W}F_0 \in \mathcal{S}_\epsilon$ is obtuse for a metric $\mathbf{G}'$ within ϵ, whose zero pattern in the frame of F is one of the 38 faces; the frame of F is one of the frames projected onto, the projection onto that face is at distance at most $\epsilon < 4\epsilon$ from $\mathbf{G}$, and F is the base superbase of its closure, so F is enumerated and passes the filter. No projection within $4\epsilon < 1$ is singular, since a singular metric differs from $\mathbf{G}$ by at least 1 in the norm along its null vector. □ The

radii are computed exactly from the balls of Lemma A6 at $\epsilon = \frac{1}{4}$.[z59]

Corollary A8 (overlap: widening one side suffices). (i) If $\|X\|_{\mathbf{A}} \le \epsilon < 1$ then $\|X\|_{\mathbf{A}+X} \le \epsilon/(1-\epsilon)$. (ii) Let $\mathbf{P}$ be an integer operator whose price measured at the new cell, $\|\mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P} - \mathbf{G}_N\|_{\mathbf{G}_N}$, is at most $\epsilon < 1$, and let F be obtuse for $\mathbf{G}_N$. Then $\mathbf{P}F \in \mathcal{S}_{\epsilon'}(\mathbf{G}_R)$ with $\epsilon' = \epsilon/(1-\epsilon)$. *Proof.* (i) With $Y = \mathbf{A}^{-1/2}X\mathbf{A}^{-1/2}$, $\|Y\|_F \le \epsilon$, the matrix $(\mathbf{A}+X)^{-1/2}X(\mathbf{A}+X)^{-1/2}$ is orthogonally similar to $(I+Y)^{-1/2}Y(I+Y)^{-1/2}$, whose eigenvalues are $y_i/(1+y_i)$ for the eigenvalues y_i of Y; since $|y_i| \le \epsilon$, $|y_i/(1+y_i)| \le |y_i|/(1-\epsilon)$, and the Frobenius norm is at most $\epsilon/(1-\epsilon)$. (ii) By (i) with $\mathbf{A} = \mathbf{G}_N$ and $X = \mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P} - \mathbf{G}_N$, the metric $\mathbf{G}_N$ is within ϵ' of $\mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P}$ in the norm at the latter. F is obtuse for $\mathbf{G}_N$; carried back by $\mathbf{P}$, which is an isometry of the norm, $\mathbf{P}F$ is obtuse for a metric within ϵ' of $\mathbf{G}_R$ in the norm at $\mathbf{G}_R$. □ The bound in (i) is attained when Y has a single non-zero eigenvalue, $-\epsilon$; it is what the implementation uses: a reference widened at $\epsilon/(1-\epsilon)$ contains the reduced superbase of every new cell whose true operator has price at most ϵ.[z60]

Proposition A9 (the price of a twofold). Let g be an integer matrix with $g^2 = I$, let $V_g = \{\mathbf{G} : g^{\mathsf{T}}\mathbf{G}g = \mathbf{G}\}$ be the metrics for which g is a symmetry, and let $\mathbf{G}^* \in V_g$. In the norm at $\mathbf{G}^*$, the price of g at $\mathbf{G}$ is twice the distance from $\mathbf{G}$ to V_g: $\|g^{\mathsf{T}}\mathbf{G}g - \mathbf{G}\|_{\mathbf{G}^*} = 2\,d_{\mathbf{G}^*}(\mathbf{G}, V_g)$. *Proof.* The map $\sigma : X \mapsto g^{\mathsf{T}}Xg$ on symmetric matrices is an involution, and it is an isometry of the inner product $\langle X, Y\rangle_{\mathbf{G}^*} = \mathrm{tr}(\mathbf{G}^{*-1}X\mathbf{G}^{*-1}Y)$ because $g\mathbf{G}^{*-1}g^{\mathsf{T}} = \mathbf{G}^{*-1}$. Its $+1$ and -1 eigenspaces are therefore orthogonal, and V_g is the $+1$ eigenspace. Write $\mathbf{G} - \mathbf{G}^* = X_+ + X_-$ in those eigenspaces. Then $g^{\mathsf{T}}\mathbf{G}g - \mathbf{G} = \sigma(\mathbf{G}) - \mathbf{G} = -2X_-$, while the nearest point of V_g to $\mathbf{G}$ is $\mathbf{G}^* + X_+$, at distance $\|X_-\|_{\mathbf{G}^*}$. □ Prices and distances measured at $\mathbf{G}$ rather than $\mathbf{G}^*$ agree with these to first order in $\|\mathbf{G} - \mathbf{G}^*\|$. The subspace of a sort to which g belongs, read in the frame in which the sort's distance is attained, is contained in $V_{g'}$ for the conjugate g' of g in that frame, so the distance to the sort is at least $d(\mathbf{G}, V_{g'})$; it is equal when

$V_{g'}$ is that subspace, as for the unique twofold of a primitive monoclinic lattice, and the constrained projection onto a sort (non-negative conorms) is at least as far as the unconstrained one. The common scale lies in V_g, so the price after the scale fit is the same to first order. The rule is checked on the worked cell, on a pseudo-tetragonal cell and along a scan of β for a monoclinic cell.[z60]

Proposition A10 (candidates form a coset). Let $\mathbf{P}_0$ be the true operator from a reference R to a new cell N, with price p_0 at $\mathbf{G}_N$; let H_ϵ be the set of integer matrices h with $\|h^{\mathsf{T}}\mathbf{G}_R h - \mathbf{G}_R\|_{\mathbf{G}_R} \le \epsilon$, and C_ϵ the candidates, the integer matrices of price at most ϵ at $\mathbf{G}_N$. To first order in the prices: (i) $H_{\epsilon-p_0}\mathbf{P}_0 \subseteq C_\epsilon \subseteq H_{\epsilon+p_0}\mathbf{P}_0$; (ii) if no self-map of R has price in $(\epsilon, 2\epsilon+c\epsilon^2]$, with c the constant of the second-order term, then H_ϵ is a group; (iii) if no self-map of R has price in $(\epsilon-p_0, 2\epsilon+c\epsilon^2]$, then $C_\epsilon = H_\epsilon\mathbf{P}_0$. *Proof.* For any integer h, $(h\mathbf{P}_0)^{\mathsf{T}}\mathbf{G}_R(h\mathbf{P}_0) - \mathbf{G}_N = \mathbf{P}_0^{\mathsf{T}}(h^{\mathsf{T}}\mathbf{G}_R h - \mathbf{G}_R)\mathbf{P}_0 + (\mathbf{P}_0^{\mathsf{T}}\mathbf{G}_R\mathbf{P}_0 - \mathbf{G}_N)$. The first term has, in the norm at $\mathbf{P}_0^{\mathsf{T}}\mathbf{G}_R\mathbf{P}_0$, the norm of $h^{\mathsf{T}}\mathbf{G}_R h - \mathbf{G}_R$ at $\mathbf{G}_R$, which is the price of h; measured at $\mathbf{G}_N$ instead it differs by a factor $1 + O(p_0)$. The triangle inequality for the norm at $\mathbf{G}_N$ then gives $|\,\mathrm{price}(h\mathbf{P}_0) - \mathrm{price}(h)\,| \le p_0$, which is (i), since every integer matrix $\mathbf{P}$ is $h\mathbf{P}_0$ with $h = \mathbf{P}\mathbf{P}_0^{-1}$. For (ii), $h_1, h_2 \in H_\epsilon$ have $\mathrm{price}(h_1 h_2) \le \mathrm{price}(h_1) + \mathrm{price}(h_2) + c\epsilon^2 \le 2\epsilon + c\epsilon^2$ by the same argument, so the gap forces $\mathrm{price}(h_1 h_2) \le \epsilon$; a finite set of matrices closed under products is a group. For (iii), the gap $(\epsilon-p_0, 2\epsilon]$ contains $(\epsilon-p_0, \epsilon+p_0]$ because $p_0 \le \epsilon$, so $H_{\epsilon+p_0} = H_{\epsilon-p_0} = H_\epsilon$ and (i) gives equality. □ The bounds hold to first order in ϵ. The implementation's prices include the common scale fit; for a unimodular h the logarithms of the eigenvalues of $\mathbf{G}_R^{-1}h^{\mathsf{T}}\mathbf{G}_R h$ sum to zero, so the trace term that the scale removes is of second order and the fit changes a price only at second order. The condition in (iii) is sufficient, not necessary: the worked cell at $\epsilon = 0.074$ has self-maps of price 0, 0.016 and 0.10, so the condition fails, yet $H_\epsilon = \{\pm I, \pm 2_y\}$ is a group, which the implementation checks directly.[lib]

Appendix B
A band on the conorms

The simplest widening of the closure is an absolute band τ in Å^2 on the conorms,

$$\mathcal{S}_\tau(\mathbf{G}) = \{\, F \;:\; \text{all six conorms of } F \text{ under } \mathbf{G} \text{ are } \geq -\tau \,\}, \tag{6}$$

Selling reduction with its boundary moved from 0 to $-\tau$. It can be computed by idealising: set every conorm of the reduced superbase F_0 that is at most 4τ to zero, enumerate the exact closure of the idealised lattice Λ^*, and keep the members whose conorms under the measured metric are all at least $-\tau$. The construction is exact when λ_1, the squared length of the shortest lattice vector, exceeds 32τ (Theorem B5), a ratio independent of the scale of the cell: for an isotropic cell it allows an angular error of about 1.8°. It is the special case of the strain norm for cells whose edges are similar in length, it is what a tolerant reduction uses implicitly, and it fails for anisotropic cells, where the band must cover the conorms between the long vectors (Section 5.3). On lattices of five zero patterns with zero conorms placed at random it was exact at every band from $\lambda_1/32$ to $\lambda_1/6$ in which the idealised lattice existed;[z55] on dense two-zero grids idealising at 2τ was already exact, at τ not.[z46] For the worked cell at $\tau = 20\,\text{Å}^2$, within the hypothesis, it returns the six members of the ring.[z45]

For a superbase F and two metrics, $\delta_F(\mathbf{G}, \mathbf{G}')$ is the largest change among F's six conorms.

Proposition B1 (characterisation superbase by superbase). $\mathcal{S}_\tau(\mathbf{G}) = \{F : F \in \mathcal{C}(\mathbf{G}')$ for some $\mathbf{G}'$ with $\delta_F(\mathbf{G}, \mathbf{G}') \leq \tau\}$. *Proof.* ($\supseteq$) If F is obtuse for $\mathbf{G}'$ and $\delta_F \leq \tau$, every conorm of F under $\mathbf{G}$ is at least $-\tau$. ($\subseteq$) The metric whose conorms in the frame of F are $\max(p_{ij}, 0)$ exceeds $\mathbf{G}$ by a positive semidefinite term, so it is positive definite and F is obtuse for it. □

Lemma B2 (class gap). Let Λ^* be obtained from F_0 by setting its conorms $\le \theta$ to zero, be positive definite, and let m^* be the smallest conorm of F_0 above θ. A superbase not obtuse for Λ^* has $e_c \ge 4m^*$ in some class. *Proof.* In the frame of F_0, Lemma A2 gives $e_c(F) = 4\sum p^0_{ij}\gamma_{ij}$, and the excess for Λ^* is the same sum over the slots above θ. If those terms vanish in all seven classes, F is obtuse for Λ^*; otherwise some $\gamma_{ij} \ge 1$ multiplies a $p^0_{ij} \ge m^*$. □

Lemma B3 (small excess in the band). If $\lambda_1 > 32\tau$ and every conorm of F is at least $-\tau$, then $e_c(F) \le 16\tau$ in every class. *Proof.* Call a slot large if $p_{ij} > 8\tau$, and let N be the negative slots. The large slots connect the four vectors, since otherwise a split crossed only by slots with $p_{ij} \le 8\tau$ has a partial sum of squared length at most $32\tau < \lambda_1$. Lemma A2 in the frame of F gives $0 \le e_c(F) = 4\sum_N |p_{ij}|\gamma_{ij} - 4\sum_{p_{ij}>0} p_{ij}\gamma_{ij}$, so $\sum_{\text{large}} p\gamma \le \tau \sum_N \gamma$. If a large γ were positive, Lemma B4(ii) would give $8\tau\sum_{\text{large}}\gamma < \tau\sum_N\gamma \le 8\tau\sum_{\text{large}}\gamma$; so the large γ vanish, and Lemma B4(i) gives $e_c \le 4\tau\cdot 4$. □

Lemma B4 (two constants on four points). Let L be a set of slots connecting the four vectors and N disjoint from L. (i) If $\gamma_{ij} = 0$ on L, then $\sum_N \gamma_{ij} \le 4$. (ii) If $\sum_L \gamma_{ij} > 0$, then $\sum_N \gamma_{ij} \le 8\sum_L \gamma_{ij}$. Both bounds are attained. *Proof.* In (i) $|\Delta k| \le 1$ on L, so all $|k_i| \le 3$, and the maximum is a finite enumeration over the 38 connected slot sets, the sets N and the 14 partial sums. In (ii) the ratio is enumerated for $|k_i| \le 15$; beyond, some $|k_i| \ge 16$, and since L connects the four vertices with $k_0 = 0$, $Q = \sum_L \Delta k^2 \ge 16^2/3 > 75$. For each pattern (L, N) let $\rho_{L,N}$ be the largest generalised eigenvalue of the quadratic form $\sum_N \Delta k^2$ against $\sum_L \Delta k^2$; the largest over all pairs (L, N) is $3 + 2\sqrt{2}$. Then $\sum_N \gamma \le \rho_{L,N} Q + \sqrt{|N|\rho_{L,N} Q}$ and $\sum_L \gamma \ge Q - \sqrt{|L|Q}$, and the ratio of these bounds, $(\rho_{L,N}\sqrt{Q} + \sqrt{|N|\rho_{L,N}})/(\sqrt{Q} - \sqrt{|L|})$, decreases in Q, so it suffices to check it at $Q = 75$ for each pattern with its own $\rho_{L,N}$, where it is below 8 in every case; with the uniform value $3 + 2\sqrt{2}$ it would not be. The lemma is computer-assisted. Equality in (ii): $L = \{01, 02, 13\}$, $N = \{03, 12, 23\}$, $s = (0, 0, 1, 1)$.

□[z44]

Theorem B5. Let F_0 be obtuse for $\mathbf{G}$, $\lambda_1 > 32\tau$, and Λ^* obtained by setting every conorm of F_0 that is at most 4τ to zero. Then Λ^* is positive definite and $\mathcal{S}_\tau(\mathbf{G}) = \{F \in \mathcal{C}(\Lambda^*) : p_{ij}(F;\mathbf{G}) \geq -\tau\}$. *Proof.* If the slots of F_0 above 4τ did not connect the four vectors, some partial sum would have squared length at most $16\tau < \lambda_1$. The inclusion $\supseteq$ is the definition. For $F \in \mathcal{S}_\tau(\mathbf{G})$, Lemma B3 gives $e_c(F;\mathbf{G}) \leq 16\tau$; the excess for Λ^* is the same sum with the zeroed slots dropped, so $e_c(F;\Lambda^*) \leq 16\tau < 4m^*$, and Lemma B2 makes F obtuse for Λ^*. □

The operators from F_0 to the members of $\mathcal{C}(\Lambda^*)$ are integer matrices independent of the metric, so one enumeration serves every band up to the one idealised at.

Lemma B6 (nesting). If $Z' \subseteq Z$ are sets of conorms of F_0 set to zero, every member of the closure for Z' is a member of the closure for Z. *Proof.* The members for Z' are reached from F_0 by words of flips, each at a slot that is zero at that point of the word (Lemma A3); a flip at a slot zero for both lattices acts on both conorm vectors as the same transposition of two slots, so after each flip the superbase is obtuse for both and the next flip is again at a slot zero for both. □[z46]

Proposition B7 (reindexing guarantee in the reference frame). Let $\mathbf{G}'_N$ be the new cell's metric written in the reference's reduced frame, $\|\mathbf{G}'_N - \mathbf{G}_R\|_F \leq \eta < \lambda_{\min}(\mathbf{G}_R)$, let F_N be its reduced superbase with basis vectors of summed squared length V, and let $\tau \geq \sqrt{2}\,\eta\,V/(\lambda_{\min}(\mathbf{G}_R) - \eta)$ with $\lambda_1(\mathbf{G}_R) > 32\tau$. Then F_N, with its true integer coefficients in the reference frame, is a member of $\mathcal{S}_\tau(\mathbf{G}_R)$, and pairing it with the new cell's reduced superbase proposes the true operator. *Proof.* With U the coordinates of F_N's basis vectors in the reference frame and $\Delta = \mathbf{G}'_N - \mathbf{G}_R$, $\sum |u_i|^2 \leq V/(\lambda_{\min} - \eta)$; in F_N's own frame the error has Frobenius norm at most $\sum |u_i|^2\,\eta$, and every conorm shift is at most $\sqrt{2}$ times that norm. The conorms of F_N under $\mathbf{G}'_N$ are non-negative, so under $\mathbf{G}_R$ they are at least $-\tau$, and Theorem B5 applies. □

For fifteen references near zero conorms of five zero patterns, each with twelve new cells carrying a random metric error and a random reindexing, the true operator was among the candidates in 180 of 180 cases at the band given by the coverage bound below; the guarantee of Proposition B7 is conservative, a median 13η against a band near 3η that was enough.[z47]

From a metric error to a band. A metric error $\Delta\mathbf{G}$ shifts the conorm of a member with coordinate vectors u_i, u_j by $-u_i^{\mathsf{T}}\Delta\mathbf{G}\,u_j$, at most $c_{ij}\eta$ with $c_{ij} = \|\operatorname{sym}(u_i u_j^{\mathsf{T}})\|_F$; every member survives every error of norm η exactly when $\tau \geq \tau_{\mathrm{cov}} = \max(c_{ij}\eta - p_{ij})$, read off the stored closure. The largest constant on zero conorms is $\sqrt{2}$ with one zero, $\sqrt{5}$ with two, and $2\sqrt{3}$ in six of the sixteen labelled cuboid frames, so $\tau = 3\eta$ is not a worst-case guarantee in every frame.[z36] These frame-dependent constants are what the strain norm removes.

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Synopsis

Two data sets from one crystal can reduce to different cells of the same lattice, related by an integer matrix that is not a lattice symmetry. All reduced cells of a lattice and the matrices between them form a finite groupoid; widened in a basis-independent strain norm, it is enumerated exactly for measured cells and gives every candidate reindexing operator within a stated tolerance, with a price for each.
